

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Title – LearnMate AI – Agentic Personalized Learning Pathway

Domain of Project – Education

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Problem Statement #12: Agentic AI for Personalized Course Pathways
AICTE – IBM University Engagement Program

Problem Statement #12 : Agentic AI for Personalized Course Pathways

The Challenge:

Modern technical education presents learners with an overwhelming abundance of fragmented online courses, videos, and tutorials without structured pedagogical guidance. Conventional learning platforms deliver static, one-size-fits-all curricula that fail to account for pre-existing proficiencies, specific career goals, and variable learning velocity. Without continuous diagnostic skill assessment, automated gap analysis, and dynamic curriculum adaptation, students suffer high attrition and persistent misalignment with industry demands.

The Objective:

Develop an autonomous Multi-Agent AI system powered by IBM Granite that diagnoses student proficiencies, pinpoints granular skill gaps against target career goals, generates ordered personalized learning pathways, and dynamically adapts the curriculum in real time based on progress and feedback:

- **Autonomous Student Assessment** – Evaluate background, interests, and declared skills to construct a precise baseline competency profile.
- **Granular Skill Gap Analysis** – Map current abilities against target goal prerequisites to isolate missing theoretical and practical skills.
- **Dynamic Roadmap Planning** – Synthesize an ordered sequence of 4–7 focused courses with modular topics and calibrated duration estimates.
- **Domain-Verified Resource Curation** – Enrich courses with vetted materials from trusted platforms (Coursera, edX, Python Docs) with zero dead links.
- **Closed-Loop Re-Act Adaptation** – Analyze completion telemetry to autonomously recalibrate roadmap pacing and structure as learners advance.

Proposed Solution: Agentic AI-Powered Personalized Course Pathways

LearnMate AI introduces a full-stack, autonomous Multi-Agent system powered by IBM Granite and IBM watsonx.ai. Rather than delivering static course catalogs, LearnMate orchestrates five specialized agents cooperating through an autonomous pipeline that personalizes, monitors, and continuously recalibrates each student's learning journey:

- 1. Student Assessment Agent** – Ingests student profile, declared background, and interests; prompts IBM Granite to evaluate baseline competence and determine appropriate foundational levels.
- 2. Skill Gap Analysis Agent** – Contrasts student profile against prerequisite concepts of the target goal, producing prioritized skill gaps and pedagogical justifications.
- 3. Roadmap Planner Agent** – Designs an ordered, multi-course curriculum (4–7 courses) tailored to fill identified gaps with clear milestone descriptions and realistic time commitments.
- 4. Learning Resource / Course Agent** – Enriches course modules with curated learning links, enforcing strict allowlist validation against 35+ verified educational domains (Coursera, edX, MIT OCW) to eliminate hallucinated URLs.
- 5. Progress & Adaptation Agent** – Monitors course completion metrics and learner feedback; executes a closed-loop Re-Act cycle to adapt and persist updated roadmap structures when pacing requires intervention.

Full-Stack Implementation – Integrated through an asynchronous Python FastAPI backend, SQLite relational database, and an intuitive React 19 + Vite frontend featuring real-time progress tracking.

Technology Used

Core Technology Stack & Architecture Components

IBM Bob Platform – Used for designing, managing project structure (.bob, agents, tools, learnmate full-stack app), automated build pipelines (Stage 3 frontend build), and orchestrating multi-agent developer workflows.

IBM Granite Foundation Model (ibm/granite-4-h-small) – Enterprise language model accessed via IBM watsonx.ai REST APIs with dynamic IAM bearer token authentication; drives multi-agent reasoning, skill gap identification, curriculum planning, and roadmap adaptation.

IBM watsonx.ai Platform – Provides high-availability inference endpoints, prompt governance, parameter tuning (greedy decoding, max tokens, repetition penalty), and structured JSON output serialization.

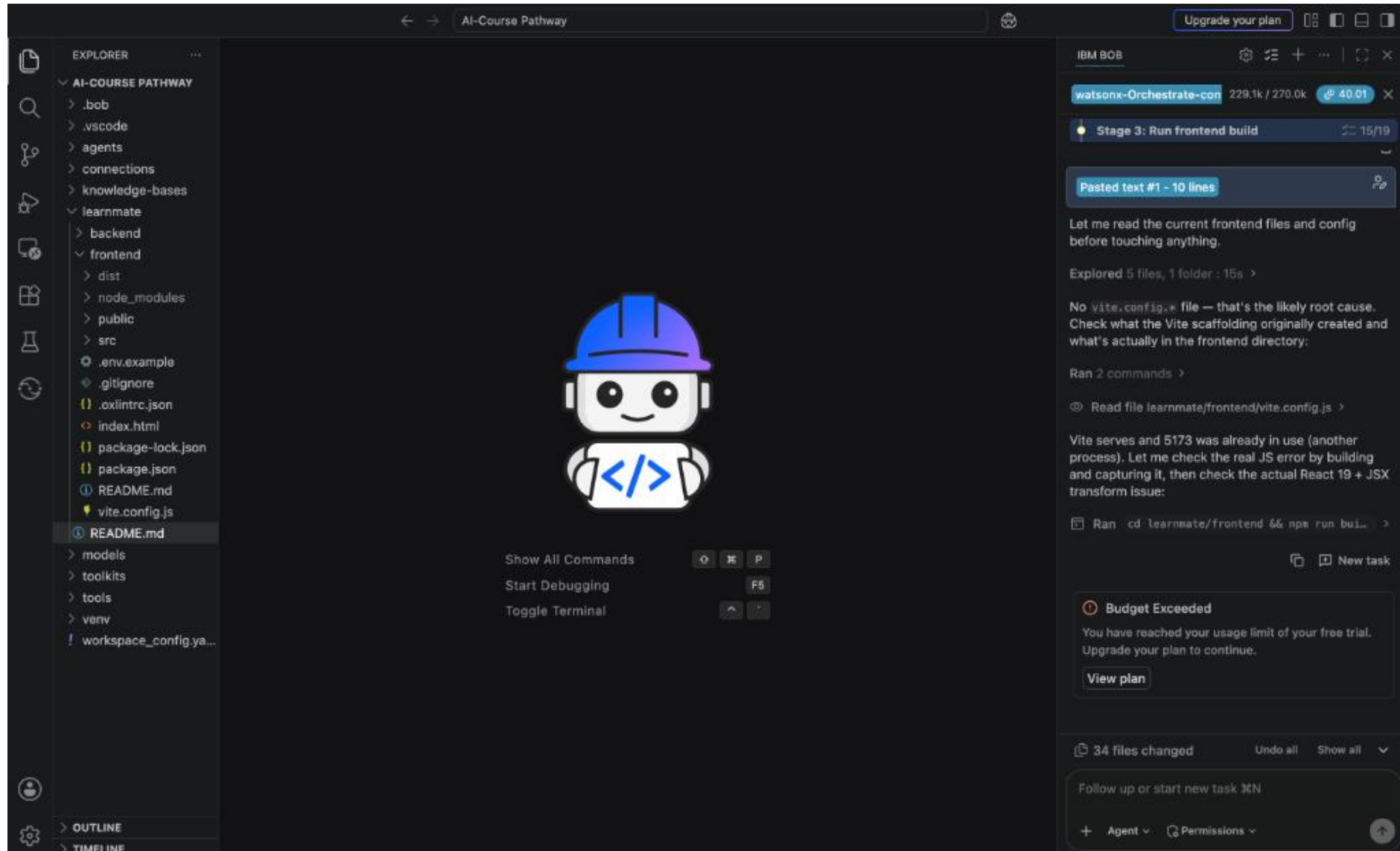
Python 3.9+ & FastAPI Backend – High-performance asynchronous REST API framework orchestrating modular agent routers, CORS middleware, lifespan lifecycle management, and automatic OpenAPI/Swagger documentation generation.

Pydantic v2 & SQLAlchemy 2.x ORM – Enforces strict input validation, strongly typed schema contracts, and relational persistence across students, skill profiles, roadmaps, course modules, and adaptation logs.

React 19 & Vite 8 SPA – Modern, responsive web interface offering a guided 3-step workflow (Profile Setup → Goal Definition → Interactive Learning Dashboard) with live progress sliders and AI adaptation triggers.

SQLite Database & Pytest Test Suite – Lightweight relational datastore preserving roadmap versions and student state, validated by 18 automated unit and integration tests across all agent endpoints.

Required Files created using IBM bob -



LearnMate UI – Step 1: Student Profile & Skill Baseline

Create your profile

Tell us about yourself so IBM Granite can personalise your learning path.

Full name

Email

Interests (comma-separated)

Experience level

Beginner

Create profile →

LearnMate UI – Step 2: Target Goal & Competency Specification

 **LearnMate AI**

Agentic Personalised Learning — IBM Granite

Backend online

+ Granite ready

✓ Profile →

2 Goal →

3 Dashboard

What do you want to learn?

← Back

Hi rohan shet — tell IBM Granite your goal.

Learning goal

Machine Learning

Your current skills (comma-separated)

python

Build my roadmap →

System Architecture Blueprint: LearnMate AI Multi-Agent System

LearnMate AI — System Architecture

IBM SkillsBuild / AICTE Problem Statement #12 · Agentic AI for Personalised Course Pathways

FastAPI 0.115 + Python 3.9 IBM Granite ibm/granite-4-h-small React 19 + Vite 8 SQLite + SQLAlchemy 2.x 18 tests passing

FRONTEND

REACT 19 SPA · VITE 8 · HTTP://LOCALHOST:5173

Step 1 — ProfileForm

name · email · interests · level

POST /api/students

Step 2 — GoalForm

learning goal · current skills

POST /api/learn

Step 3 — Dashboard

assessment · skill gaps · roadmap · resources · progress sliders

POST /api/progress

POST /api/roadmap/adapt

Header

backend status · Granite badge · step indicator

GET /api/health

API base: VITE_API_URL (default http://localhost:8000/api) · All API keys kept server-side only

HTTP/JSON · CORS (localhost:5173, :3000)

API LAYER

FASTAPI 0.115 · UVICORN · HTTP://LOCALHOST:8000 · PYDANTIC V2 VALIDATION

GET /api/health

POST /api/students

GET /api/students/{id}

POST /api/assessment

POST /api/roadmap

GET /api/roadmap/{student_id}

POST /api/roadmap/adapt ★

POST /api/progress

POST /api/learn ★★

★ adapt: persists Roadmap.status = "adapted" + timestamps adaptation note when should_adapt=true · ★★ learn: full 5-agent orchestration pipeline

Agent dispatch · async/await

ORCHESTRATION

POST /API/LEARN PIPELINE — SEQUENTIAL AGENT EXECUTION

LearnOrchestrator

app/api/learn.py

Step 1 AssessmentAgent → upsert skills

Step 2 SkillGapAgent

Step 3 RoadmapPlannerAgent → write roadmaps + courses

Step 4 ResourceAgent

Step 5 ProgressAgent (baseline)

Returns LearnResponse (14 fields)

+

AdaptOrchestrator

app/api/roadmap.py → /adapt

Input progress_data · feedback · roadmap_id

Step 1 ProgressAgent (with real progress)

Step 2 if should_adapt → DB update

Roadmap.description += note

Roadmap.status = "adapted"

Returns AdaptRoadmapResponse (7 fields)

Role of Agentic AI in the Solution

Autonomous Multi-Agent Collaboration in LearnMate AI

Autonomous Workflow Orchestration – Instead of an isolated chatbot, five specialized agents coordinate sequentially. Assessment diagnoses baseline competence, SkillGap isolates missing prerequisites, Planner constructs modular curricula, Resource curates validated materials, and Progress adapts pathways.

Cognitive Specialization & Modularity – Each agent operates with tailored system prompts and targeted IBM Granite inference calls, preventing context dilution and ensuring high-fidelity outputs for complex academic planning.

Domain Guardrails & URL Validation – The Resource Agent actively enforces strict platform validation across 35+ verified educational platforms, sanitizing any invented URLs to guarantee safe, authentic student resources.

Closed-Loop Re-Act Adaptation Loop – Learner progress telemetry feeds directly into the Progress Agent. If a student struggles or requests adjustments, IBM Granite dynamically refactors course ordering and pacing.

State Persistence & Longitudinal Memory – Agents persist structured JSON payloads into relational SQLite schemas, maintaining an evolving academic history that informs future curriculum adjustments.

Token Usage

IBM Bob Platform Token Usage: 40.18 Credits & 229.1k Context Tokens Consumed across Agentic Workflows

After the Prompt

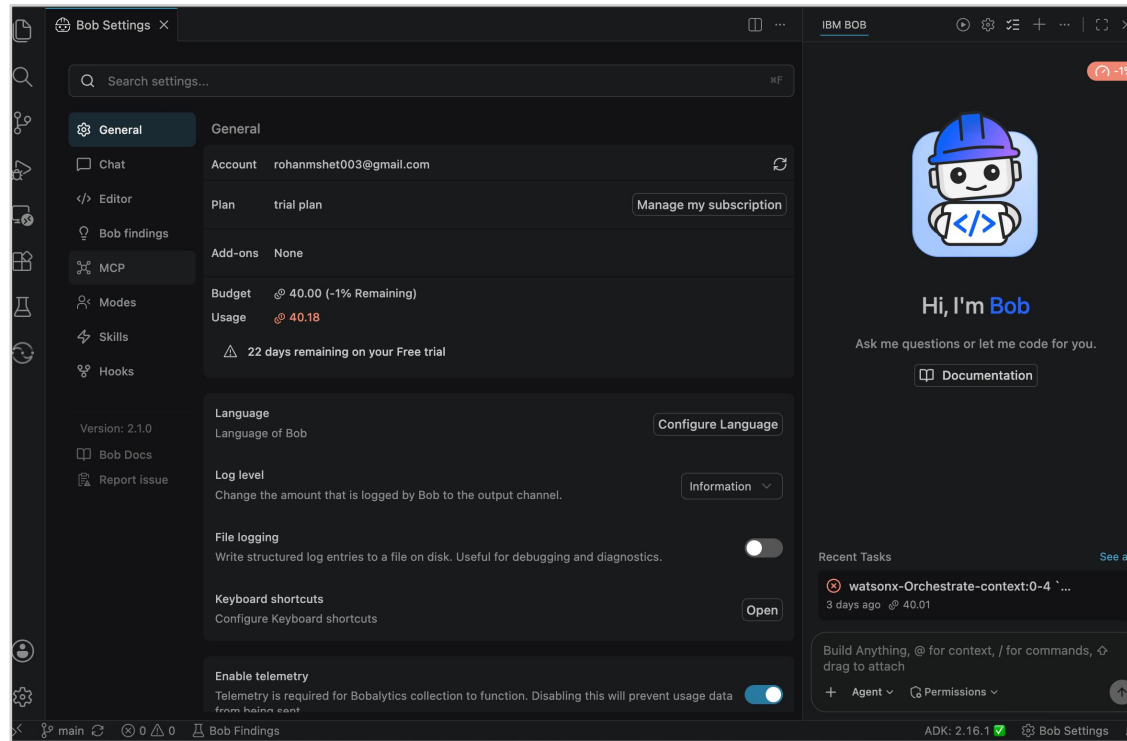


Figure 5: IBM Bob Settings & Usage telemetry confirming 40.18 credits consumed (-1% remaining) after task execution.

Project Output: Real-Time Multi-Agent Pipeline Execution



Agentic Personalised Learning — IBM Granite

Backend online + Granite ready

✓ Profile → 2 Goal → 3 Dashboard



IBM Granite is building your personalised roadmap...

Running 5 AI agents: Assessment → Skill Gap → Roadmap →
Resources → Progress

Project Output: Synthesized Personalized Learning Roadmap



Agentic Personalised Learning — IBM Granite

Backend online

+ Granite ready

✓ Profile → ✓ Goal → 3 Dashboard

Machine Learning Foundations

rohan shet · beginner level + IBM Granite

Start over

Overall progress

0 of 7 courses complete · 0% overall

Assessment

SUMMARY

The student has a foundation in Python programming, which is essential for machine learning. However, they will need to develop knowledge in statistics, linear algebra, probability, data preprocessing, model evaluation, and various machine learning algorithms to progress towards their goal.

DETECTED SKILLS

python

SKILL GAPS

statistics linear algebra probability data preprocessing
model evaluation machine learning algorithms

Skill gap analysis

PRIORITY SKILLS TO DEVELOP

statistics linear algebra

EXPLANATION

Machine learning heavily relies on statistical concepts and linear algebra for

Your learning roadmap

#1 Statistics for Machine Learning

Learn the fundamental statistical concepts needed for machine learning, including descriptive statistics, probability distributions,...

beginner 10h

0% ☐

#2 Linear Algebra Essentials

Master the linear algebra concepts essential for machine learning, such as vectors, matrices, matrix operations, eigenvalues, and eigenvectors.

beginner 12h

0% ☐

#3 Probability Theory Basics

Understand the probability theory fundamentals required for machine learning, including random variables, probability distributions, Bayes'...

Project Output: Granular Modules, Verified Resources & Adaptation

 **LearnMate AI**

Agentic Personalised Learning — IBM Granite

Backend online

+ Granite ready

✓ Profile →

✓ Goal →

3 Dashboard

thinking, which is essential for machine learning.

Open resource →

Statistical Inference

edX

This beginner-level course focuses on the principles of statistical inference, including estimation, hypothesis testing, and confidence intervals. It covers the mathematical foundations and practical applications of statistical inference, which are crucial for understanding and evaluating machine learning models.

Open resource →

Probability and Statistics

Udacity

This comprehensive beginner-friendly course covers the fundamentals of probability theory and statistical methods. It includes topics such as probability distributions, random variables, hypothesis testing, and regression analysis. The course provides a solid foundation for understanding the statistical concepts used in machine learning.

Open resource →

Statistics Fundamentals

DataCamp

This beginner-level course introduces the core concepts of statistics, including descriptive statistics, probability, sampling, and hypothesis testing. It provides a gentle introduction to statistical thinking and helps learners develop the skills needed to analyze and interpret data in the context of machine learning.

Open resource →

#6

Fundamental machine Learning Algorithms

Explore the core machine learning algorithms, such as linear regression, logistic regression, k-nearest neighbors, decision trees,...

intermediate 18h

0%

#7

Machine Learning Foundations Review

Review and consolidate the key concepts, techniques, and algorithms covered in the previous courses to reinforce your understanding of...

beginner 5h

0%

AI adaptation

Update your course progress above, then ask IBM Granite to analyse your progress and recommend adjustments to your roadmap.

+ Analyse & adapt roadmap

Novelty and Uniqueness

LearnMate AI stands out by combining Multi-Agent AI, IBM Granite, and closed-loop personalization:

Multi-Agent Pedagogical Pipeline – Decomposes educational planning across 5 cooperating agents rather than relying on a single hallucination-prone prompt.

Closed-Loop Re-Act Adaptation – Continuously responds to completion telemetry and learner feedback to autonomously restructure and reprioritize course roadmaps.

Domain-Verified Resource Curation – Enforces programmatic domain guards against 35+ verified educational sources (Coursera, edX, MIT OCW), eliminating invalid URLs.

Granular Diagnostic Assessment – Performs semantic gap analysis between declared competencies and target learning goals to calibrate optimal starting levels.

Enterprise AI Governance – Powered by IBM Granite on watsonx.ai with dynamic IAM bearer token authentication, greedy decoding, and safe fallback handling.

Full-Stack Production Readiness – A responsive React 19 frontend integrated with a high-performance Python FastAPI backend and SQLite relational persistence.

Git Hub Link

Public GitHub Repository:

<https://github.com/rohanmshet003-rora/learnmate-ai>

Repository Structure & Verified Contents: The repository contains the complete, production-ready source code for LearnMate AI:

- 1. FastAPI Application Backend** – FastAPI app, modular agent routers (app/agents/), ORM models (app/models/), Pydantic schemas, and Granite async service client.
- 2. React 19 Frontend Application** – Vite 8 Single Page Application (frontend/src/App.jsx) with responsive CSS design system and 3-step interactive user journey.
- 3. Automated Test Suite** – Pytest automated test suite (tests/test_agents.py) containing 18 unit and integration tests verifying all 5 agents and the full /api/learn pipeline.
- 4. Comprehensive Documentation** – Complete README.md with architecture overview, API specifications, environment setup instructions, and execution workflows.

Future Scope

Strategic Roadmap & Future Enhancements

- 1. Multi-Modal Educational Content Ingestion** – Extend Assessment and Resource agents to ingest university course syllabi (PDFs), video lecture transcripts, and academic transcripts to align customized pathways with university degree requirements.
- 2. Autonomous Adaptive Quizzing & Skill Verification** – Introduce an automated testing agent that synthesizes dynamic diagnostic coding quizzes and conceptual assessments, verifying mastery before unlocking advanced curriculum stages.
- 3. Deep LMS Integration (Moodle, Canvas, Blackboard)** – Implement LTI (Learning Tools Interoperability) compliance to embed LearnMate AI directly within institutional Learning Management Systems, offering faculty real-time telemetry on student progress and AI adaptations.
- 4. Collaborative Peer Learning & Mentorship Matching** – Leverage vector embeddings of learner goals and progress to connect students pursuing complementary learning pathways, fostering automated study groups and peer code reviews.

Certificates Earned:

IBM SkillsBuild Credly Certificate: Getting Started with Artificial Intelligence
Student: Rohan M Shet | Issued: Aug 30, 2026 | Verify: <https://www.credly.com/badges/6fa73995-d517-4ef2-8ab0-aa6a3b562d16>



Certificates Earned:

IBM SkillsBuild Course Completion: Lab: Troubleshoot Your Code Using IBM Bob
Student: Rohan M Shet | Course ID: ALM-COURSE_4071307 | Completed: 01 Sep 2026

IBM SkillsBuild

Completion Certificate



This certificate is presented to

Rohan M Shet

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

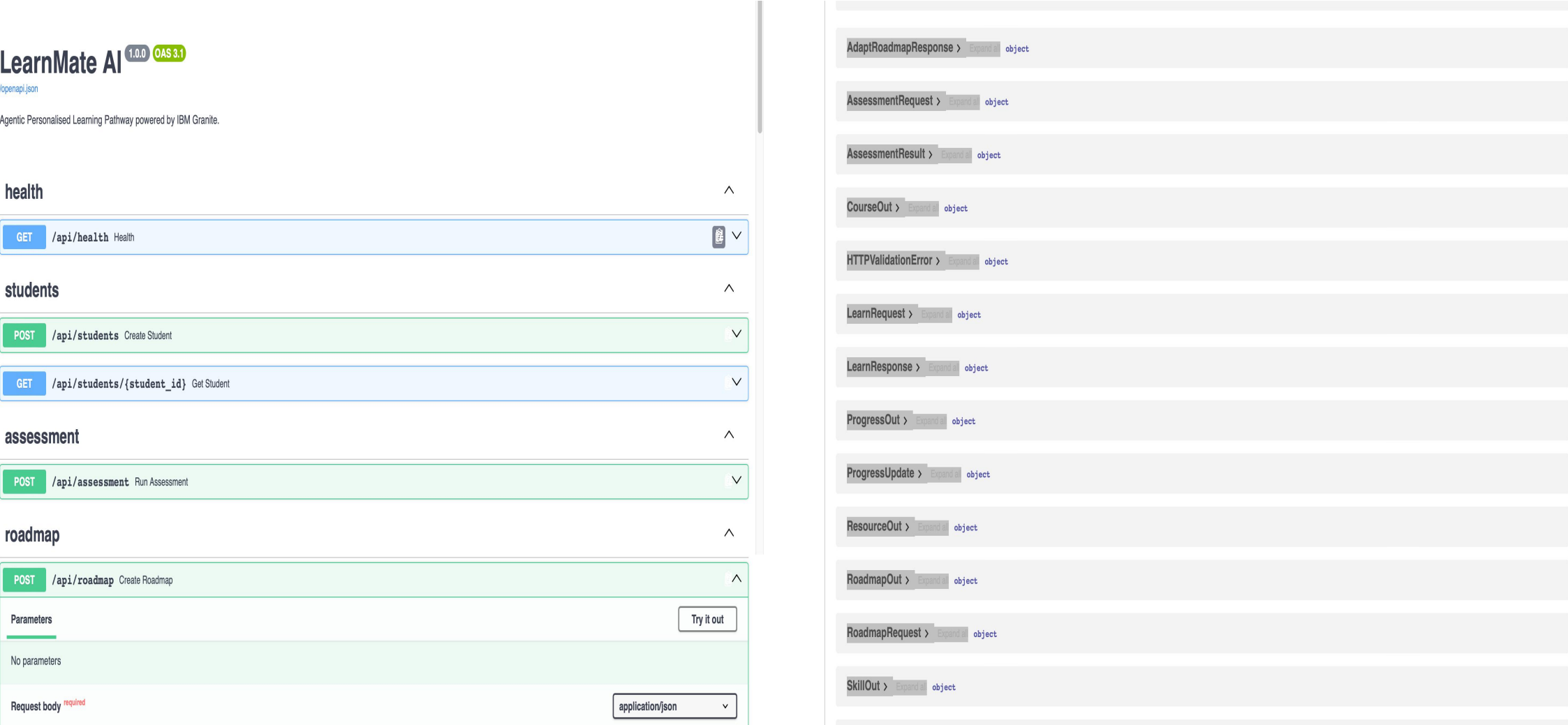
According to the Adobe Learning Manager system of record

Completion date: 01 Sep 2026 (GMT)

Learning hours: 30 mins

Project Backend APIs & Data Contracts

Figure 8: FastAPI RESTful API Endpoints (Left) & Pydantic Data Contract Specifications (Right) for LearnMate AI.



Thank You!

Thank you for your time and interest.